

STP

S1(config)#
S1(config)#spanning-tree vlan 1 priority 0
S1(config)#end

S1 (Root Bridge)

Switch1

Fa0/1

Fa0/2

Switch3

Fa0/1

Fa0/2

Switch2

Fa0/1

Fa0/2

S2 (Backup Root)

S3 (Path Manipulation)

S2 (Backup Root)

S2(config)#
S2(config)#spanning-tree vlan 1 priority 4096
S2(config)#end
S2#

S3(config)#interface fa0/1
S3(config-if)#spanning-tree cost 50
S3(config-if)#end

Medium-Level STP Question (Exam Style)

Topology Scenario

3 Switches: S1, S2, S3

All connected in triangle:

S1 ↔ S2

S2 ↔ S3

S3 ↔ S1

VLAN: VLAN 1

Default priority on all switches: 32768

? Question

Perform the following tasks:

Task 1: Root Bridge Selection

Set S1 as Root Bridge for VLAN 1

Task 2: Secondary Root

Configure S2 as Backup Root Bridge

Task 3: Port Cost Manipulation

On S3, make the link towards S1 less preferred (so traffic prefers path via S2)

Task 5: Verification

Verify:

Root bridge

Port roles (Root/Designated/Blocking)

S1#show spanning-tree vlan 1

VLAN0001

Spanning tree enabled protocol ieee

Root ID Priority 1

Address 0040.0BB7.21BB

This bridge is the root

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 1 (priority 0 sys-id-ext 1)

Address 0040.0BB7.21BB

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Aging Time 20

Interface	Role	Sts	Cost	Prio.Nbr	Type
Fa0/2	Desg	FWD	19	128.2	P2p
Fa0/1	Desg	FWD	19	128.1	P2p

S1#